

Analysis note: charged-particle multi-particle cumulants in ALEPH hadronic e^+e^- collisions at the Z pole

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Abstract

This note documents a first charged-particle multi-particle cumulant analysis of archived ALEPH hadronic e^+e^- collisions at $\sqrt{s} \simeq 91.2$ GeV. The analysis follows the public Electron-Positron Alliance note style: the main body defines the observable, data setup, event and track selections, cumulant calculation, statistical uncertainty treatment, and the current preliminary results. The reported observables are $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ as functions of selected charged-particle multiplicity, evaluated with respect to both the laboratory beam axis and the event thrust axis. The cumulant implementation uses exact ordered, distinct-particle correlations through eighth order and propagates statistical uncertainties with delete-one-chunk jackknife resampling over sixteen event chunks. The present result is a detector-level charged-particle study; detector-response, acceptance, and model-dependent systematic uncertainties are not yet included. A 1994 reconstructed-MC baseline is included for detector-level comparisons of the beam-axis and thrust-axis trends, including rapidity-gap and two-subevent cumulant constructions.

Keywords: ALEPH, archived e^+e^- data, charged-particle cumulants, flow harmonics, Q-cumulants

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1 Introduction

Long-range azimuthal correlations and multi-particle cumulants are central tools for separating collective and non-collective correlation structures in high-energy collisions. In hadronic and nuclear collisions, the hierarchy of $v_n\{2\}$, $v_n\{4\}$, $v_n\{6\}$, and $v_n\{8\}$ provides a compact way to test whether the measured azimuthal anisotropy is dominated by a common event-level geometry or by few-particle correlations. The same techniques can be applied to archived e^+e^- data, where the absence of beam remnants and a well-defined color-singlet initial state provide a clean reference for jet fragmentation, parton-shower, and hadronization effects.

The starting point for this analysis is the ALEPH StudyMult charged-particle workflow and the previous ALEPH two-particle-correlation analysis of archived LEP data [1]. The present note extends that infrastructure from two-particle Fourier coefficients to exact multi-particle cumulants through eight particles. The implementation follows the Q-cumulant formalism of Refs. [2, 3] and the cumulant combinations used in small-system flow measurements, including the CMS high-multiplicity pp reference analysis [4].

This document intentionally follows the compact public-note format used in recent Electron-Positron Alliance analysis notes. The main body contains the observable definition, active data setup, analysis method, validation, and preliminary results, while detailed corrections and systematic variations are identified as future work rather than folded into the present detector-level result.

2 Observable Definition

For each selected event, charged particles are represented by their azimuthal angles ϕ_i in a chosen coordinate system. The present study uses two coordinate definitions:

- beam axis: ϕ_i is the laboratory azimuth around the ALEPH beam axis;
- thrust axis: the event is rotated into a frame defined by the event thrust axis, and ϕ_i is the azimuth around that axis.

The harmonic used in this note is $n = 2$. For an event with multiplicity M , the ordered distinct-particle $2k$ -particle correlation is

$$\langle 2k \rangle_n = \frac{1}{M(M-1)\cdots(M-2k+1)} \sum_{\substack{i_1, \dots, i_{2k} \\ \text{all distinct}}} \exp[in(\phi_{i_1} + \cdots + \phi_{i_k} - \phi_{i_{k+1}} - \cdots - \phi_{i_{2k}})]. \quad (1)$$

The event-ensemble averages of these quantities are accumulated with their natural ordered-tuple weights. The cumulants are then

$$c_2\{2\} = \langle 2 \rangle, \quad (2)$$

$$c_2\{4\} = \langle 4 \rangle - 2 \langle 2 \rangle^2, \quad (3)$$

$$c_2\{6\} = \langle 6 \rangle - 9 \langle 4 \rangle \langle 2 \rangle + 12 \langle 2 \rangle^3, \quad (4)$$

$$c_2\{8\} = \langle 8 \rangle - 16 \langle 6 \rangle \langle 2 \rangle - 18 \langle 4 \rangle^2 + 144 \langle 4 \rangle \langle 2 \rangle^2 - 144 \langle 2 \rangle^4. \quad (5)$$

The corresponding flow-like coefficients are defined as

$$v_2\{2\} = \sqrt{c_2\{2\}}, \quad (6)$$

$$v_2\{4\} = [-c_2\{4\}]^{1/4}, \quad (7)$$

$$v_2\{6\} = [c_2\{6\}/4]^{1/6}, \quad (8)$$

$$v_2\{8\} = [-c_2\{8\}/33]^{1/8}. \quad (9)$$

If the required cumulant sign is not physical for a given bin, the corresponding $v_2\{m\}$ point is suppressed in the plotted result and is stored as zero in the exported CSV table.

3 Data Setup and Selections

The figures in this note currently use the merged LEP1 1992–1995 reconstructed data sample

`/mnt/data3/data3/yjlee/StudyMultSamples/ALEPH/LEP1/LEP1Merged_20200611.root.`

This file contains 3,050,610 entries in the tree `t`; all entries pass the current charged-particle selection and multiplicity-bin requirement. The input format is the StudyMult fixed-array layout with `year`, `RunNo`, `EventNo`, `nParticle`, `px`, `py`, `pz`, and `pwflag` branches. The year composition is 551,474 events from 1992, 538,601 from 1993, 1,365,440 from 1994, and 595,095 from 1995. A converted 1991 ROOT sample was not found in the available merged samples, so this production uses the available merged LEP1 statistics from 1992–1995 rather than the complete 1991–1995 data set.

Charged particles are selected using the StudyMult charged-particle convention

$$\text{pwflag} \in \{0, 1, 2\}, \quad (10)$$

with $p_T > 0.4$ GeV for both the beam-axis and thrust-axis cumulants. The event thrust axis is computed from reconstructed particle momenta, following the local ALEPH StudyMult convention used by the prototype analyses. The charged-particle multiplicity bins used in this note are

$$[0, 10), [10, 15), [15, 20), [20, 25), [25, 30), [30, 35), [35, 40), [40, \infty).$$

The corresponding event counts are shown in Table 1. All beam-axis and thrust-axis cumulant figures are binned on this laboratory selected charged-particle multiplicity; the thrust-selected multiplicity is retained as a diagnostic but is not used as the horizontal-axis binning in the present production.

Table 1: Selected event counts in the multiplicity bins used for the current detector-level cumulant analysis.

Multiplicity bin	Events
[0, 10)	241919
[10, 15)	1046419
[15, 20)	1132225
[20, 25)	500481
[25, 30)	113682
[30, 35)	14738
[35, 40)	1095
[40, ∞)	51

4 Cumulant Implementation

The analysis code computes the ordered correlations in Eq. (1) using an exact elementary-symmetric polynomial recursion over the event-level phase factors $\exp(in\phi_i)$. This avoids self-correlations by construction and uses the same ordered-particle denominator as the Q-cumulant definition. The recursion is applied for 2, 4, 6, and 8 particles in each multiplicity bin and for both axis definitions.

The implementation was checked in two ways. First, the code contains a `--SelfTest 1` mode that compares the production correlator against brute-force ordered distinct-particle sums for low-multiplicity toy events. Second, the cumulant combinations and $v_2\{m\}$ normalizations were compared against the published Q-cumulant formulae in Refs. [2, 3, 4]. The signs and normalization factors in the cumulant and $v_2\{m\}$ definitions above match those references.

5 Statistical Uncertainties

The production job was split into sixteen contiguous event chunks. Each chunk stores independent numerator and denominator sums for every correlator and multiplicity bin. The summary step first merges all chunks

to obtain the central value and then recomputes the full nonlinear cumulant chain after removing one chunk at a time. The quoted statistical uncertainty is the delete-one-chunk jackknife estimate

$$\sigma_{\text{JK}} = \sqrt{\frac{N_{\text{chunk}} - 1}{N_{\text{chunk}}} \sum_{j=1}^{N_{\text{chunk}}} (x_{(j)} - \bar{x})^2}, \quad (11)$$

where $x_{(j)}$ is the result with chunk j removed. This propagates the nonlinear transformations from raw correlations to cumulants and then to $v_2\{m\}$. If any leave-one sample has the wrong cumulant sign for a real-valued $v_2\{m\}$, the corresponding $v_2\{m\}$ point is suppressed rather than assigned a conditional error from only the sign-valid samples. The procedure assumes the chunks are statistically independent; if the input file ordering tracks detector running conditions, the jackknife may also contain a small run-block component.

6 Preliminary Results

Figures 1 and 2 show the current detector-level $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ results versus charged-particle multiplicity. In both coordinate systems the two-particle value is larger than the higher-order values, as expected when few-particle and jet-like correlations contribute. The higher-order beam-axis points remain numerically stable through most bins, while some high-order thrust-axis and high-multiplicity points become sign-unstable or carry large jackknife uncertainties because the event counts and ordered eight-particle statistics are limited in those bins.

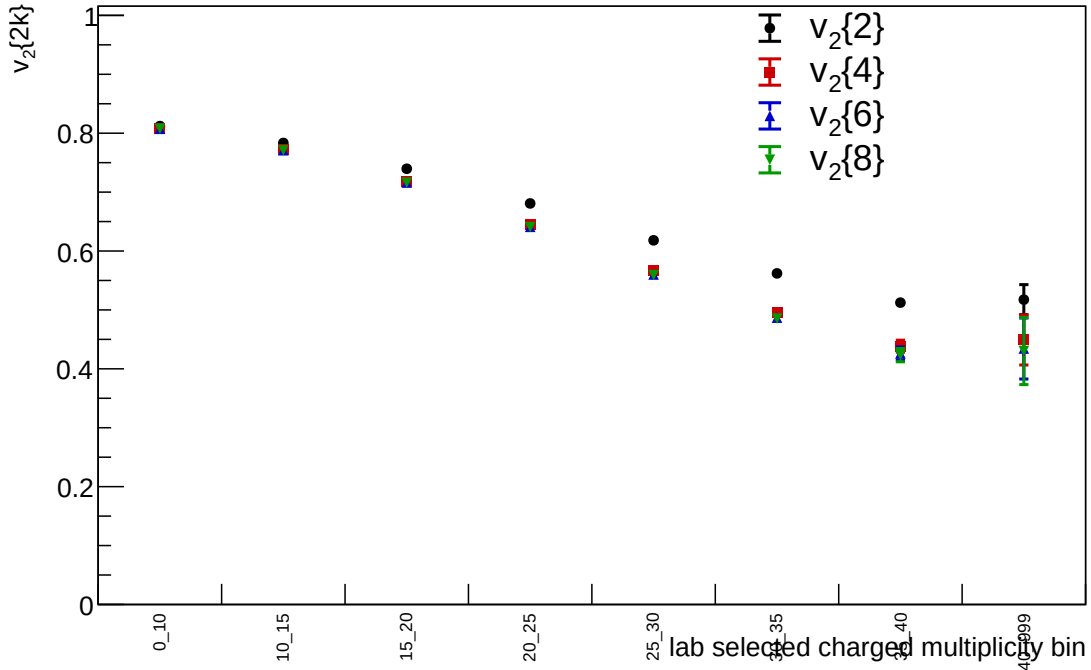


Figure 1: Detector-level charged-particle $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ as functions of selected charged-particle multiplicity using the laboratory beam-axis azimuth. Error bars are delete-one-chunk jackknife statistical uncertainties from sixteen event chunks. Missing high-order points indicate cumulant signs that do not give a real-valued $v_2\{m\}$ in the current sample.

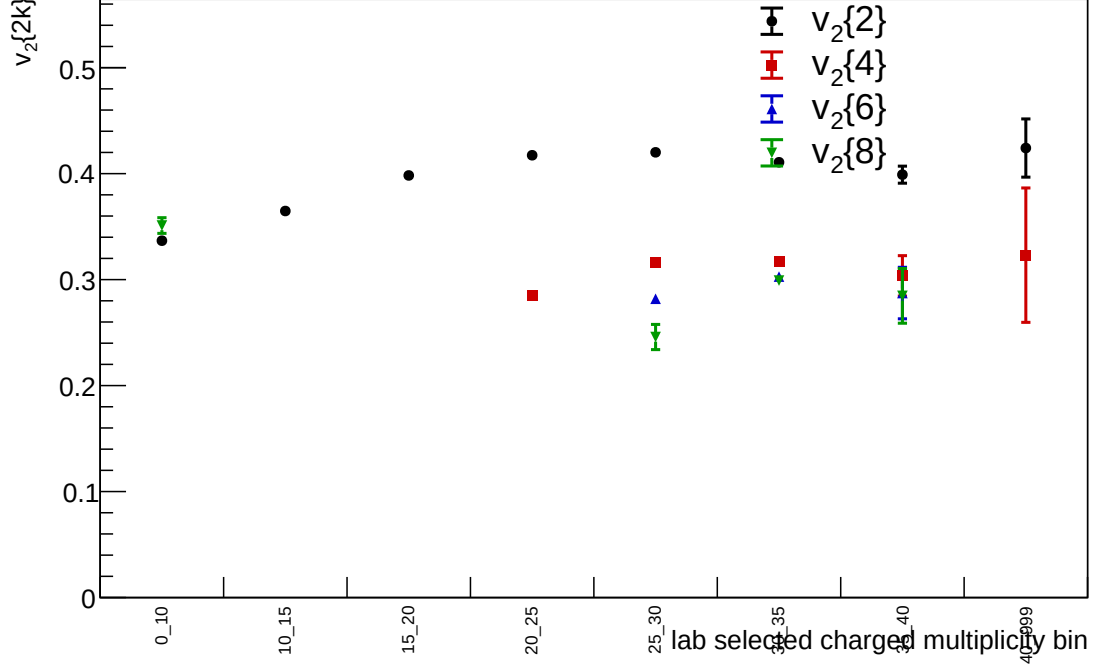


Figure 2: Detector-level charged-particle $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ as functions of selected charged-particle multiplicity using azimuth around the event thrust axis. Error bars are delete-one-chunk jackknife statistical uncertainties from sixteen event chunks.

Table 2: Numerical beam-axis $v_2\{m\}$ values. Uncertainties are jackknife statistical uncertainties. Entries shown as “—” are sign-invalid or unavailable in the current sample.

Multiplicity	$v_2\{2\}$	$v_2\{4\}$	$v_2\{6\}$	$v_2\{8\}$
[0, 10)	0.8121 ± 0.0004	0.8078 ± 0.0004	0.8075 ± 0.0004	0.8075 ± 0.0004
[10, 15)	0.7834 ± 0.0002	0.7720 ± 0.0002	0.7711 ± 0.0002	0.7709 ± 0.0002
[15, 20)	0.7396 ± 0.0002	0.7183 ± 0.0002	0.7162 ± 0.0003	0.7159 ± 0.0003
[20, 25)	0.6808 ± 0.0002	0.6449 ± 0.0003	0.6411 ± 0.0003	0.6404 ± 0.0003
[25, 30)	0.6181 ± 0.0005	0.5662 ± 0.0006	0.5602 ± 0.0007	0.5591 ± 0.0007
[30, 35)	0.5620 ± 0.0016	0.4952 ± 0.0022	0.4869 ± 0.0023	0.4852 ± 0.0023
[35, 40)	0.5124 ± 0.0077	0.4384 ± 0.0102	0.4262 ± 0.0110	0.4233 ± 0.0113
[40, ∞)	0.5174 ± 0.0255	0.4491 ± 0.0426	0.4345 ± 0.0518	0.4302 ± 0.0570

Table 3: Numerical thrust-axis $v_2\{m\}$ values. Uncertainties are jackknife statistical uncertainties. Entries shown as “—” are sign-invalid or unavailable in the current sample.

Multiplicity	$v_2\{2\}$	$v_2\{4\}$	$v_2\{6\}$	$v_2\{8\}$
[0, 10)	0.3367 ± 0.0016	—	—	0.3510 ± 0.0074
[10, 15)	0.3647 ± 0.0003	—	—	—
[15, 20)	0.3983 ± 0.0004	—	—	—
[20, 25)	0.4173 ± 0.0003	0.2846 ± 0.0015	—	—
[25, 30)	0.4201 ± 0.0006	0.3165 ± 0.0015	0.2819 ± 0.0037	0.2458 ± 0.0119
[30, 35)	0.4107 ± 0.0023	0.3168 ± 0.0037	0.3028 ± 0.0034	0.2991 ± 0.0035
[35, 40)	0.3990 ± 0.0080	0.3036 ± 0.0190	0.2873 ± 0.0243	0.2847 ± 0.0258
[40, ∞)	0.4242 ± 0.0275	0.3231 ± 0.0634	—	—

6.1 Matched 1994 reconstructed-MC comparison

A separate MC production was run on the available LEP1 1994 reconstructed-MC sample

`/mnt/data3/data3/yjlee/StudyMultSamples/LEP1MCMerged.root.`

This MC file contains 771,597 reconstructed entries in tree `t`, all from 1994. It is therefore used as a detector-level baseline rather than a strict year-matched comparison to the 1992–1995 data sample. After the charged-particle selection and multiplicity-bin requirement, 749,082 MC events enter the inclusive cumulant histograms.

Figures 3 and 4 show the standalone MC result for the beam and thrust axes. The beam-axis MC follows the same broad multiplicity dependence as the data-only result: the cumulant hierarchy is tight at low multiplicity and decreases with increasing multiplicity. In the thrust axis, the two-particle coefficient rises from low to intermediate multiplicity, while several low-multiplicity higher-order coefficients are sign-invalid and therefore not plotted.

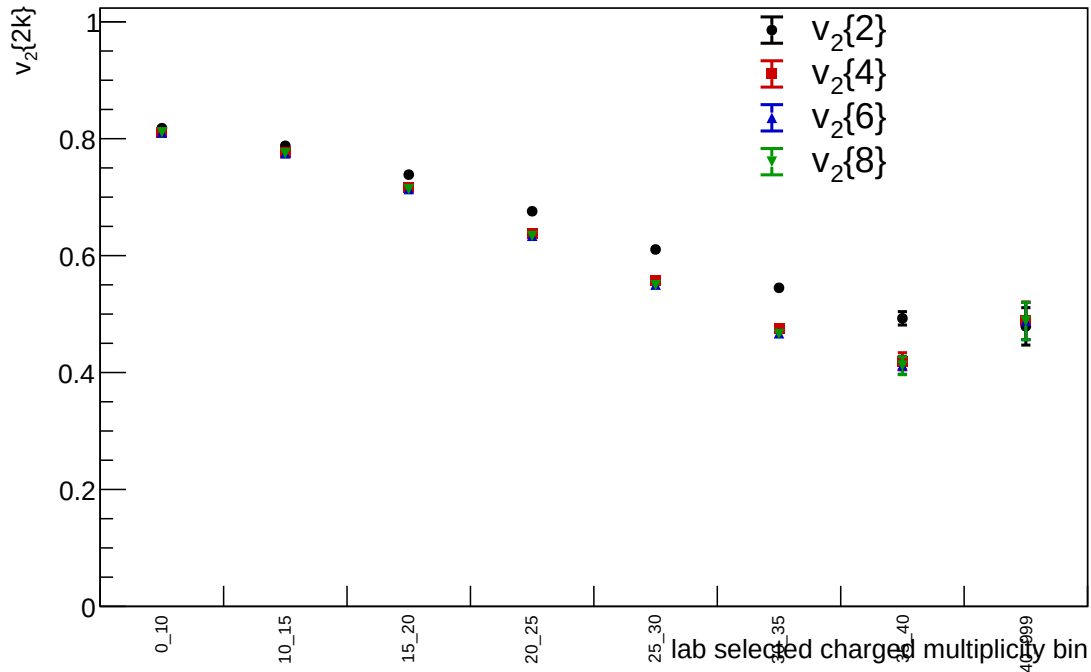


Figure 3: LEP1 1994 reconstructed-MC $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ versus selected charged-particle multiplicity using the laboratory beam-axis azimuth.

Figures 5 and 6 compare the LEP1 1992–1995 data and 1994 reconstructed MC directly. For the beam axis, the data/MC agreement is at the percent level over most of the populated multiplicity range; the largest deviations appear in the highest-multiplicity bin, where the statistical uncertainty is also largest. For the thrust axis, $v_2\{2\}$ is well reproduced through intermediate multiplicity. The higher-order thrust-axis coefficients are more sensitive to cumulant sign changes and limited statistics, so missing points indicate bins where either data or MC did not yield a real value under the standard cumulant sign convention.

6.2 Two-particle rapidity-gap comparison

The inclusive two-particle result contains every ordered charged-particle pair in an event. To test the sensitivity to short-range and same-jet correlations, the LEP1 1992–1995 data and 1994 reconstructed-MC

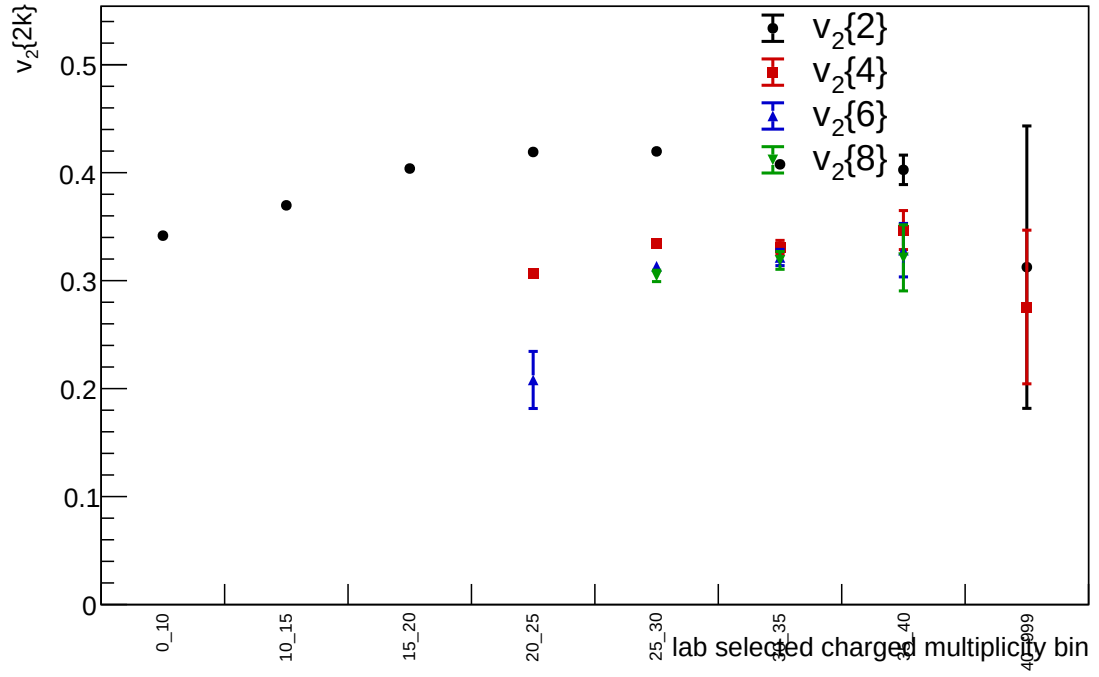


Figure 4: LEP1 1994 reconstructed-MC $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ versus selected charged-particle multiplicity using azimuth around the event thrust axis.

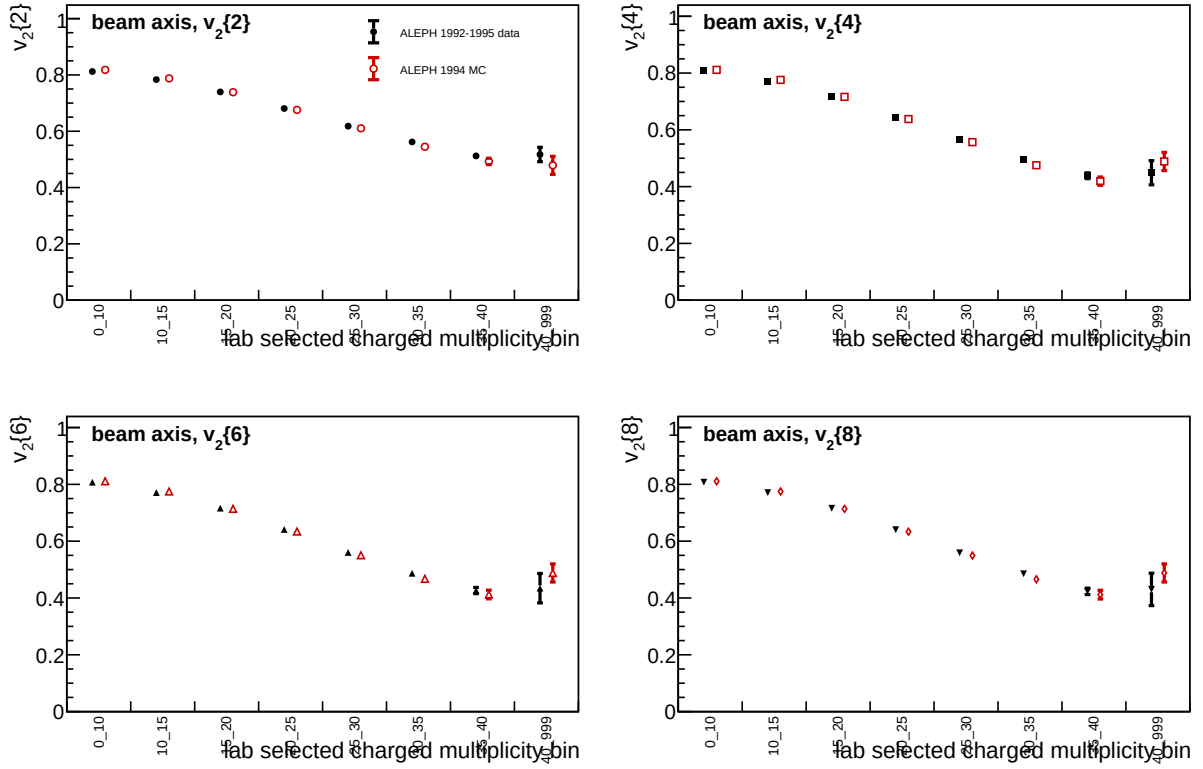


Figure 5: Beam-axis comparison of LEP1 1992–1995 data and 1994 reconstructed MC. Each panel shows one cumulant order, with data and MC offset slightly in the lab-selected charged-multiplicity bin for readability.

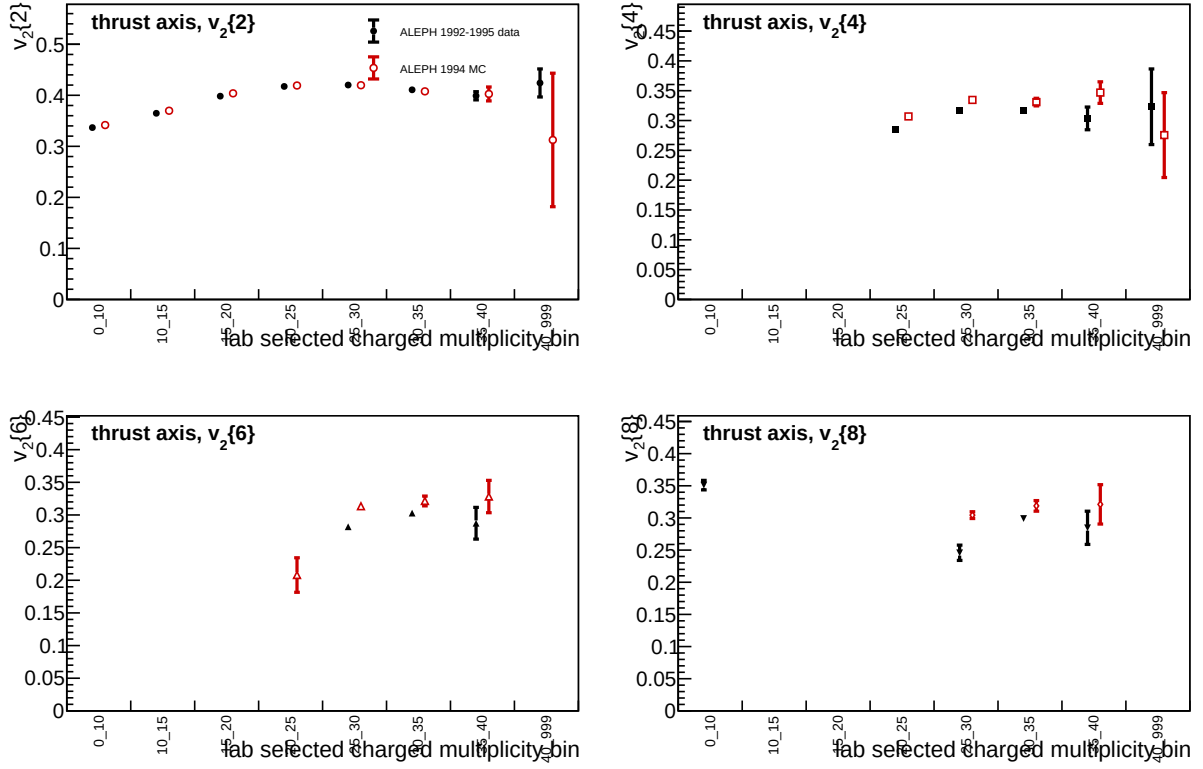


Figure 6: Thrust-axis comparison of LEP1 1992–1995 data and 1994 reconstructed MC. Missing high-order points correspond to sign-invalid cumulants or bins without a valid real-valued $v_2\{m\}$.

samples were reprocessed with an additional two-particle rapidity-gap observable. The track selection, axis definitions, charged-multiplicity bins, and sixteen event chunks are the same as in the inclusive data/MC comparison. The default additional requirement is a strict pair cut

$$|\eta_i - \eta_j| > 2.0. \quad (12)$$

A second production is also run with $|\eta_i - \eta_j| > 1.6$ to test the stability of the rapidity-gap suppression against the gap threshold. In both cases, η is evaluated in the same coordinate system as the azimuthal angle: laboratory pseudorapidity for the beam-axis result and pseudorapidity around the reconstructed thrust axis for the thrust-axis result. The horizontal axis remains the selected laboratory charged-particle multiplicity, matching the other figures in this note.

For every event and multiplicity bin, the analyzer stores the mergeable sums

$$N_2^{\Delta\eta>2} = \sum_{\text{events}} \sum_{i \neq j} \Theta(|\eta_i - \eta_j| - 2.0) \cos[2(\phi_i - \phi_j)], \quad (13)$$

$$D_2^{\Delta\eta>2} = \sum_{\text{events}} \sum_{i \neq j} \Theta(|\eta_i - \eta_j| - 2.0), \quad (14)$$

with ordered-pair weights. Equivalently, the implementation loops over unordered pairs and adds both ordered partners through a factor of two in the numerator and denominator. The gap correlation and flow coefficient are then computed only in the summary step,

$$\langle 2 \rangle_{|\Delta\eta|>2} = \frac{N_2^{\Delta\eta>2}}{D_2^{\Delta\eta>2}}, \quad v_2\{2, |\Delta\eta| > 2\} = \sqrt{\langle 2 \rangle_{|\Delta\eta|>2}}, \quad (15)$$

when the correlation is non-negative. The inclusive $v_2\{2\}$ and the rapidity-gap $v_2\{2, |\Delta\eta| > 2\}$ are both recomputed for each leave-one-chunk jackknife sample, so the plotted uncertainties include the nonlinear square-root transformation. The lower-panel gap/inclusive ratio is also formed in each leave-one sample and jackknifed as a ratio, preserving the statistical covariance between the inclusive and rapidity-gap measurements.

Figures 7 and 8 compare the inclusive and rapidity-gap measurements. MC is drawn as histogram bands and data as points. In the beam axis, the rapidity-gap value is close to the inclusive value at low multiplicity and becomes smaller from roughly 15 charged tracks onward; the data gap/inclusive ratio falls to about 0.65 in the 30–35 bin, with the high-multiplicity tail limited by statistics. The reconstructed MC follows the same trend. In the thrust axis, the rapidity-gap suppression is stronger: the data gap/inclusive ratio decreases from about 0.71 in the 0–10 bin to about 0.43 around 30–35 charged tracks, again with consistent MC behavior within the quoted statistical uncertainties. This pattern is consistent with a sizable contribution to the inclusive two-particle coefficient from short-range or jet-like correlations that are reduced by the large $|\Delta\eta|$ separation.

Figures 9 and 10 show the corresponding comparison for $|\Delta\eta| > 1.6$. As expected, the looser gap admits more event pairs and therefore gives a less suppressed gap/inclusive ratio than the $|\Delta\eta| > 2$ selection, while preserving the same qualitative multiplicity dependence.

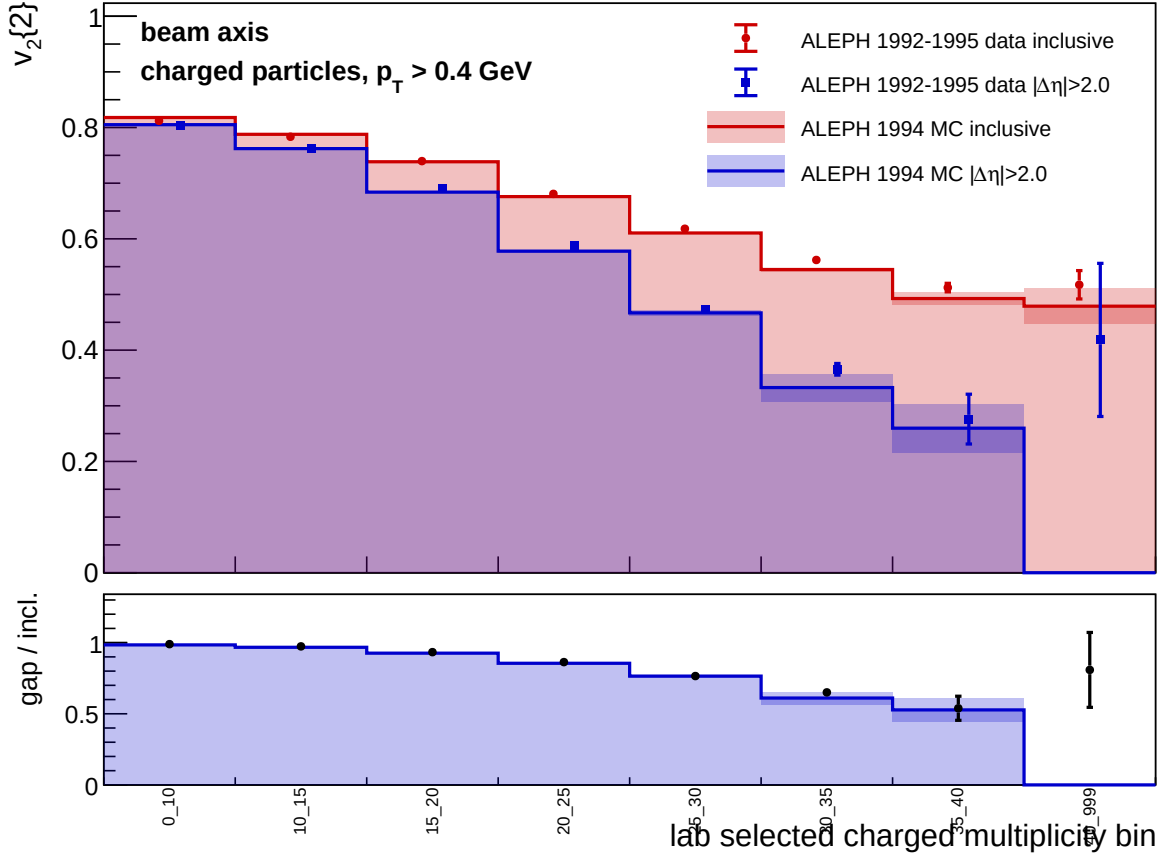


Figure 7: Beam-axis comparison of inclusive $v_2\{2\}$ and $v_2\{2, |\Delta\eta| > 2\}$ for LEP1 1992–1995 data and 1994 reconstructed MC. MC is shown as histogram bands and data as points. The lower panel shows the rapidity-gap value divided by the inclusive value.

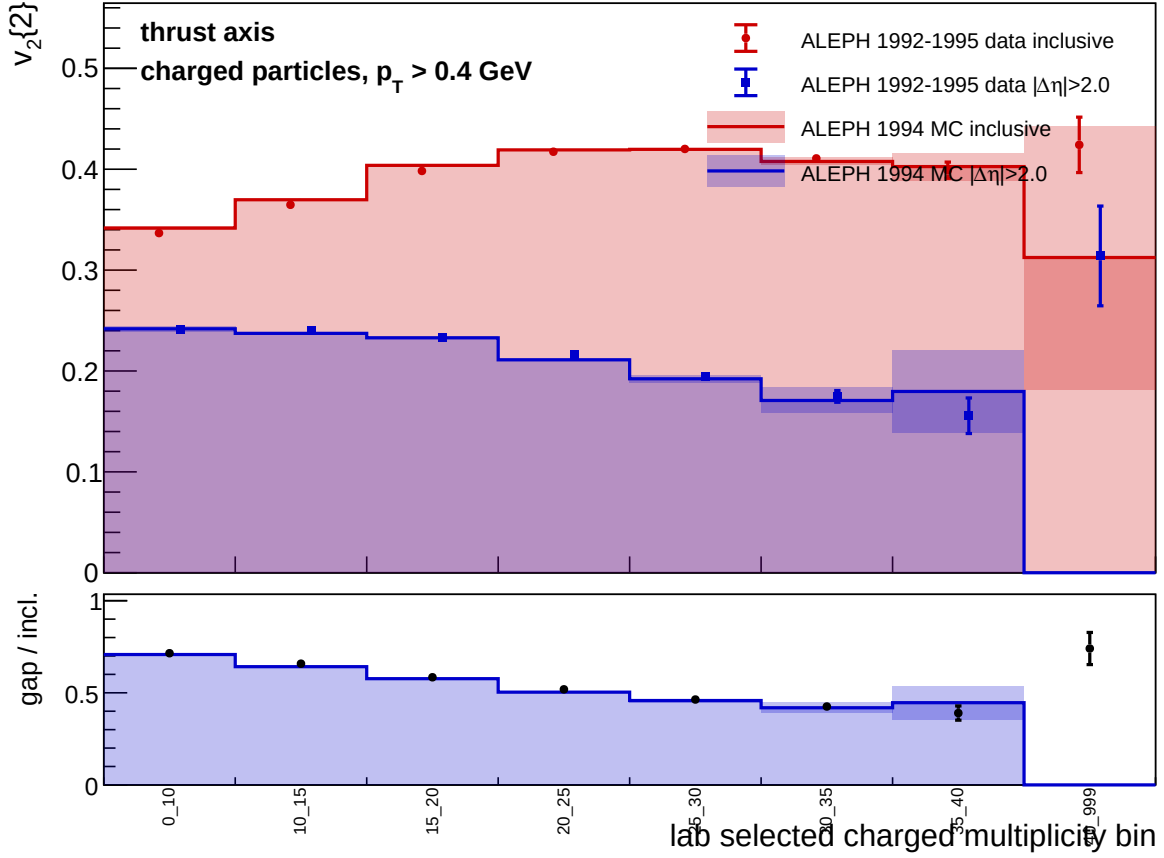


Figure 8: Thrust-axis comparison of inclusive $v_2\{2\}$ and $v_2\{2, |\Delta\eta| > 2\}$ for LEP1 1992–1995 data and 1994 reconstructed MC. The rapidity gap is evaluated using pseudorapidity around the thrust axis; the multiplicity bin remains the selected laboratory charged-particle multiplicity.

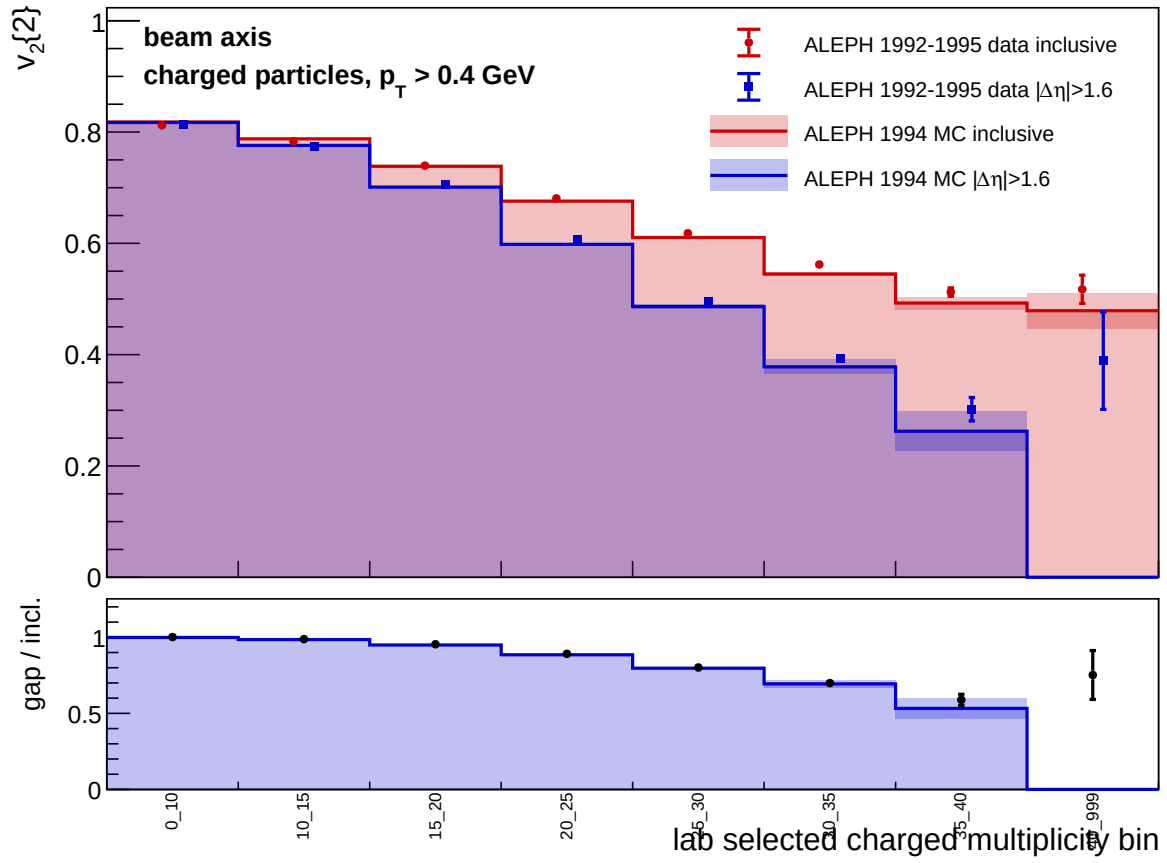


Figure 9: Beam-axis comparison of inclusive $v_2\{2\}$ and $v_2\{2, |\Delta\eta| > 1.6\}$ for LEP1 1992–1995 data and 1994 reconstructed MC.

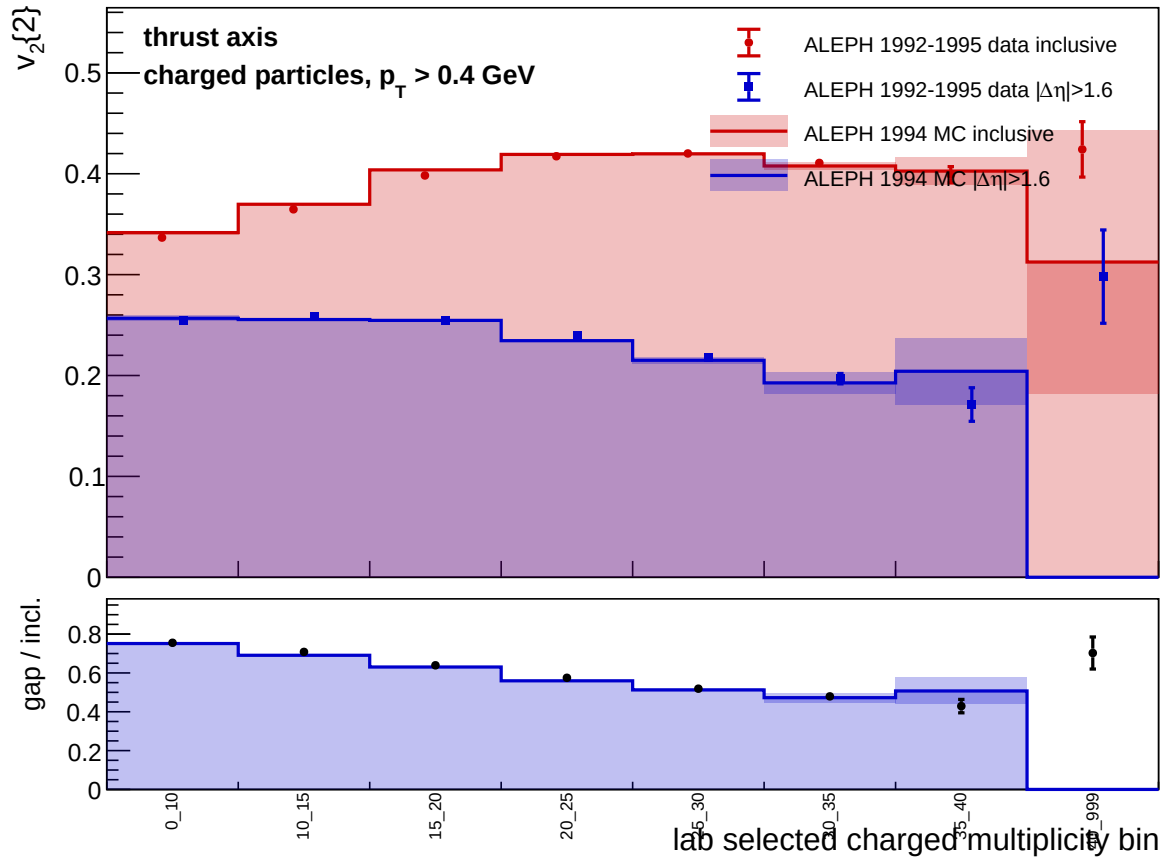


Figure 10: Thrust-axis comparison of inclusive $v_2\{2\}$ and $v_2\{2, |\Delta\eta| > 1.6\}$ for LEP1 1992–1995 data and 1994 reconstructed MC.

Two-subevent cumulant definition. For each axis frame, charged particles are split into subevent A with $\eta < -\eta_{\text{gap}}$ and subevent B with $\eta > \eta_{\text{gap}}$. The result shown here uses $\eta_{\text{gap}} = 0$, i.e. a two-hemisphere split rather than a finite exclusion gap. The $2k$ -particle correlation samples k distinct particles from A and k distinct particles from B,

$$\langle 2k \rangle_{2\text{sub}} = \left\langle e^{in(\phi_1^A + \dots + \phi_k^A - \phi_1^B - \dots - \phi_k^B)} \right\rangle,$$

with ordered tuple weights $M_A(M_A - 1) \dots (M_A - k + 1) M_B(M_B - 1) \dots (M_B - k + 1)$. The same cumulant combinations used for the all-particle result are then applied to $\langle 2 \rangle_{2\text{sub}}$, $\langle 4 \rangle_{2\text{sub}}$, $\langle 6 \rangle_{2\text{sub}}$, and $\langle 8 \rangle_{2\text{sub}}$. $v_2\{2k\}_{2\text{sub}}$ is plotted only when the corresponding cumulant has the real-valued sign convention.

6.3 Two-subevent detector-level data/MC comparison

The LEP1 1992–1995 data and 1994 reconstructed-MC samples were reprocessed with the same track selection, multiplicity bins, and sixteen-chunk jackknife procedure, adding the two-subevent cumulant sums described above. Figures 11 and 12 show MC as histograms with statistical bands and data as points with jackknife uncertainties.

For the beam axis, the two-subevent result is stable and the data/MC ratio is near unity over the populated range. The ratio is about 0.988–1.016 for $v_2\{2\}_{2\text{sub}}$ through 20–25 charged tracks and remains within the larger statistical uncertainties in the high-multiplicity tail. The higher-order beam-axis cumulants show the same qualitative behavior, with increasing statistical spread above 30 charged tracks.

For the thrust axis, $v_2\{2\}_{2\text{sub}}$ is broadly reproduced by the MC in intermediate multiplicity bins, but the higher-order two-subevent cumulants are statistically fragile. Several bins are sign-invalid in either data or MC, and valid high-order points have large jackknife uncertainties. These thrust-axis higher-order points should therefore be treated as a diagnostic of the current sample size and subevent definition, not as a corrected physics comparison.

7 Current Limitations

The present note should be read as a detector-level analysis milestone, not as a final corrected physics result. The main limitations are:

- no detector-response or acceptance correction is applied to the cumulants;
- the current reconstructed-MC comparison is a baseline check, not a detector-response correction or generator systematic uncertainty;
- no variation of the track p_T threshold, charged-particle definition, thrust-axis construction, or event-selection definition is included yet;
- high-order cumulants in the highest-multiplicity bins are statistically fragile and can become sign-invalid.

These items define the next analysis steps before the result can be interpreted as a corrected ALEPH measurement.

8 Summary

A first ALEPH charged-particle cumulant workflow has been built from the local StudyMult code base and validated against exact brute-force ordered-particle sums. The current figures use the merged LEP1 1992–1995 reconstructed data sample, with 3,050,610 selected events in the analysis multiplicity bins. The analysis extracts $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ versus charged-particle multiplicity for both the beam-axis

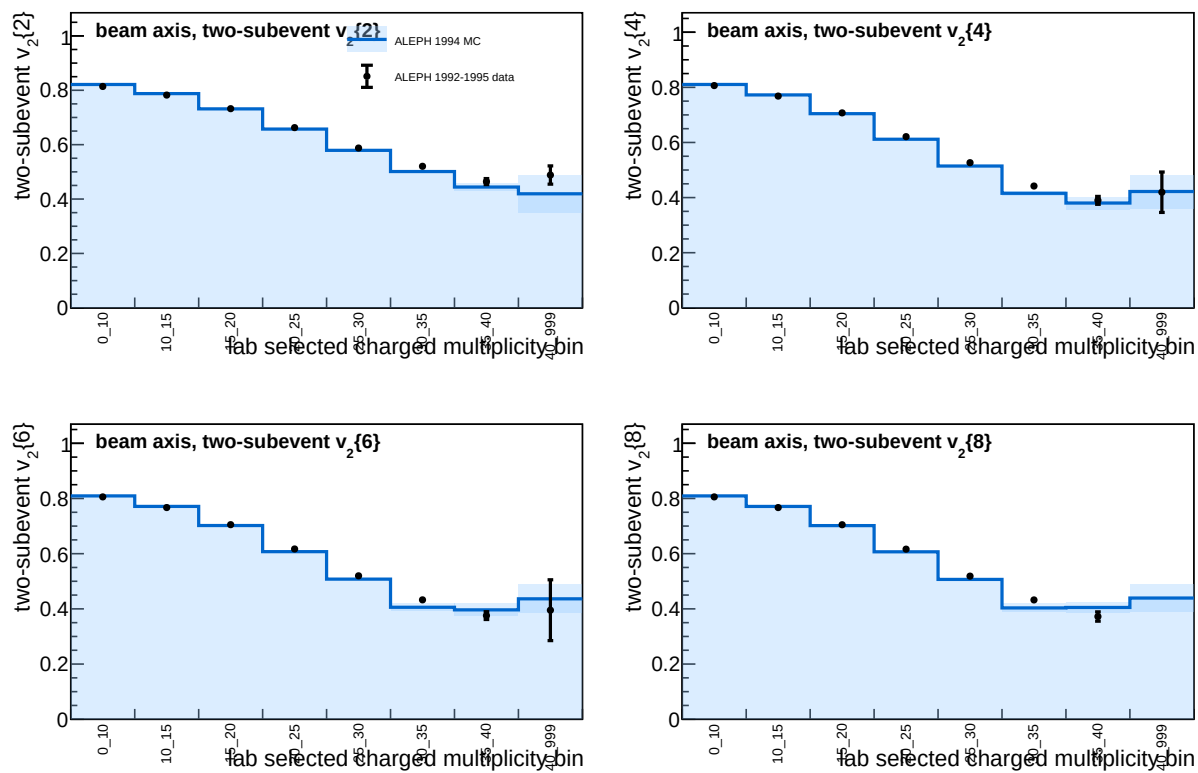


Figure 11: Beam-axis two-subevent comparison of LEP1 1992–1995 data and 1994 reconstructed MC. MC is drawn as a histogram with statistical band, while data are shown as points with delete-one-chunk jackknife uncertainties.

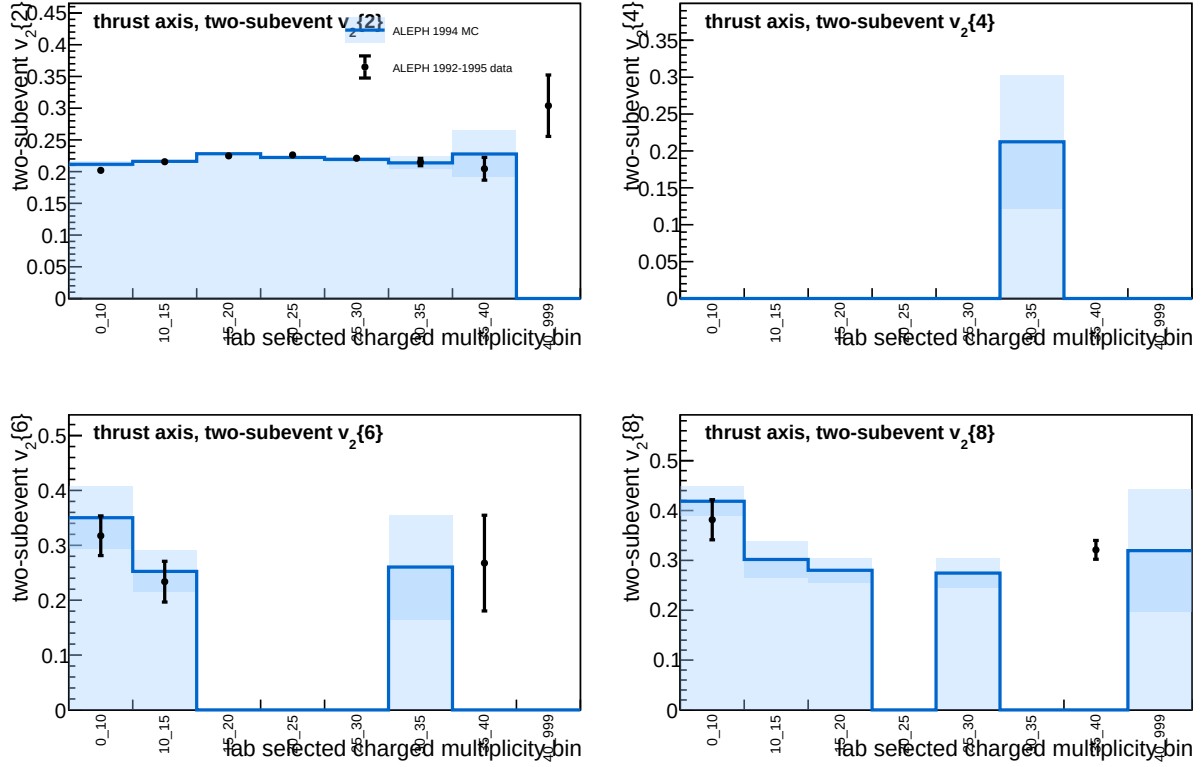


Figure 12: Thrust-axis two-subevent comparison of LEP1 1992–1995 data and 1994 reconstructed MC. Missing or zero high-order entries correspond to sign-invalid cumulants or bins without a valid real-valued $v_2\{2k\}_{2\text{sub}}$.

and thrust-axis azimuthal definitions, and compares inclusive $v_2\{2\}$ with rapidity-gap selections $|\Delta\eta| > 2.0$ and $|\Delta\eta| > 1.6$. Statistical uncertainties are propagated through the nonlinear cumulant definitions with a sixteen-chunk jackknife. The beam-axis and thrust-axis figures included in this note provide the current detector-level baseline for subsequent correction, 1991-data recovery, and systematic-uncertainty studies.

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